Set Theory: Assignment I

August 20, 2025

Q1. Show the following.

(i) $A \times B \neq B \times A$.

(ii) $A \times B = B \times A$ if and only if $A = B$, where $A \neq \emptyset,\ B \neq \emptyset$.

Q2. Consider the inverse $R^{-1} \subseteq B \times A$ of a relation $R \subseteq A \times B$, and the composition $R \circ S \subseteq A \times C$ of relations $R \subseteq A \times B,\ S \subseteq B \times C$. Show that $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$.

Q3. Prove the proposition that

(i) every partition of a set A induces an equivalence relation on A, and

(ii) conversely, any equivalence relation on A induces a partition of A.

Q4. Consider a function $f : X \to Y$. Prove the following.

(i) If $g : Y \to X$ and $gf = id_X$, the identity function on the set X, f must be one-one, and g onto.

(ii) $f(A \cap B) = f(A) \cap f(B)$ for all $A, B \subseteq X$, if and only if f is one-one.

(iii) $Y \setminus f(A) \subseteq f(X \setminus A)$ for all $A \subseteq X$, if and only if f is onto.

(iv) For any $B \subseteq Y$, $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$.

Q5. Consider functions $f : X \to Y, g : Y \to X$. Prove the following.

(i) If $gf : X \to X$ is onto, g is onto.

(ii) If $gf : X \to X$ is one-one, f is one-one.

Q6. (i) Suppose A is finite and $f : A \to A$. Prove that f is one-one if and only if f is onto.

(ii) A set X is countable if and only if there is a countable set Y and an onto map $f : Y \to X$.